

Md Mehedi Hasan

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Education

MS.c in Computer Science and Engineering

Jan 2023 – Present

Master's by Research

Title: Code quality of LLM-generated code and developers' understanding

Shahjalal University of Science and Technology, Sylhet

Bachelor of Science in Software Engineering

Feb 2018 – Jan 2023

CGPA: 3.84/4.00 - [3rd/50]

Graduated with distinction

Shahjalal University of Science and Technology, Sylhet

Relevant Courses: Data Structures and Algorithms, Introduction to Software Engineering, Software Requirement Engineering, Software Architecture and Design Patterns, Software Verification and Validation, Data Science, Artificial Intelligence, Machine Learning, Deep Learning

Research Experience

Distilling Small Language Models for Real-World Tasks Using Real and Synthetic Data Jul 2025 - Present

Code Refactoring

This thesis will demonstrate the feasibility of using small, distilled LLMs for critical software engineering tasks such as program repair and bug detection. By combining real datasets With synthetic, teacher-generated data, the work aims to bridge the gap between accuracy and efficiency, enabling practical deployment of SE-focused LLMs in low-resource environments.

Code quality of LLM generated code and developer's understanding

Oct 2025 - Present

Code quality

Investigating technical debt introduced by LLMs through comparative analysis of AI-generated vs. handwritten code quality, developer code ownership, and bug localization efficiency to assess long-term maintainability implications. investigating actionable steps to improve developers' understanding and ownership of LLM-generated code.

Teaching Experience

Shahjalal University of Science and Technology, Sylhet

Lecturer, Department of Computer Science and Engineering

2023 – Present

Sylhet, Bangladesh

Over the years, I have been teaching:

- Deep Learning, Cloud Computing, Structured Programming Language
- Software Project Management, Discrete Mathematics
- Computer Networking, Introduction to Python programming
- Web Technologies

Industry Experience

Daricomma.com

Consultant

Jun 2023 – Present

Sylhet, Bangladesh

- Deployed the system on AWS using EC2, Docker, RDS, ROUTE53
- Configured Prometheus and Grafana for monitoring
- Implemented CI/CD

Vivasoft Limited

Software Engineer L-1

Feb 2023 – Oct 2023

Banani, Dhaka

- Implemented CI/CD on TeamCity, Gitlab, and Bamboo
- Built AMI with Packer and integrated with CI/CD
- Provisioned cloud infrastructure on AWS and Huawei Cloud using Terraform, enabling automated and consistent resource management.
- Executed a seamless migration from Bitbucket to GitLab, including repository transfer and CI/CD pipeline setup.
- Set up and managed a self-hosted Git server along with Mattermost for team collaboration at Vivasoft Limited.
- Configured a monitoring system using Prometheus, Grafana, and Alertmanager to send resource alerts to the Mattermost channel.

Vivasoft Limited

Mar 2022 – Jan 2023

Jr. Devops Engineer

Banani, Dhaka

- Deployed microservices on AWS EKS using Helm for simplified Kubernetes management.
- Hosted the front-end application on AWS S3 and delivered it globally with AWS CloudFront CDN.
- Implemented end-to-end CI/CD pipelines with AWS CodePipeline, automating build, test, and deploy stages.
- Configured AWS Route 53 for DNS management and traffic routing.

Vivasoft Limited

Oct 2021 – Feb 2022

Trainee DevOps Engineer

Banani, Dhaka

- Deployed a location tracking service on AWS using ECS and Auto Scaling groups.
- Implemented CI/CD using Jenkins.
- Managed infra with Terraform.
- Set up monitoring and alerting with Prometheus and Grafana

Technical skills

DevOps tools	Docker, Docker Swarm, Kubernetes, Helm, Terraform, Consul, Tsung, Git, Packer, Ansible
Programming Languages	C, C++, Java, Go, Bash, Python
Amazon Web Services	EC2, AWS code pipeline, S3, EKS, route53, Cloudfont, VPC, ECS, ECR, ELB, RDS, DynamoDB
CI/CD & VCS	Jenkins, TeamCity, Gitlab, Bamboo, Github
Monitoring	Prometheus, Grafana
Exploring	LLM, RAG, AGENTIC AI

Projects

SWE Society Portal

Jan 2021 – Feb 2021

Express, React, Postgres

- Developed a web portal to manage profiles, events, notices, blogs, and committees for the Software Engineering Society at SUST.
- Designed a user-friendly interface for connecting to the network server and for uploading/downloading files.

File sharing system

Jun 2023 – Present

Java, Socket

- Developed a client-server application using Java where the client and server communicate via sockets.
- Enabled file operations, including upload, download, and deletion, through client-side socket commands.
- Implemented robust socket connection management, ensuring reliable data transmission between client and server.

8 Ball pool

Jul 2019 – Dec 2019

Java, LibGdx

- Develop 8 Ball pool game from scratch using LibGdx graphics library.
- Implemented game mechanics, physics, and user interface in Java with LibGDX for cross-platform compatibility.

Bus Tracker

Mar 2022 – Jan 2023

Java, GPS module

- Developed a real-time bus tracking Android application using Java, enabling live location updates and route monitoring for users.

Certification

Coursera

May 2020 – Oct 2020

Online

- Neural Networks and Deep Learning.
- Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization.
- Structuring Machine Learning Projects.
- Convolutional Neural Networks.
- Image Classification with CNNs using Keras.
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning

KodeKloud

Online

- Grafana Loki
- Certified Kubernetes Application Developer (CKAD)
- Helm for Beginners
- Hashicorp Certified Terraform Associate

Volunteer Experience

TECHNOVENT

Feb 2023

Event Coordinator

SUST

- Managed a national Hackathon event with 120+ participants
- Helped to organize the Technovent event
- Managed and led a team of 15 volunteers

ACM Student Chapter

Feb 2020 – Dec 2020

Operational Executive

SUST

- Helped the executive committee with various programs.

Awards

University Merit Stipend

Shahajalal University of Science & Technology
(SUST) 2020

National Higher Secondary School Merit Scholarship

by Government of Bangladesh 2018

Referees

Mohammad Shahidur Rahman

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Computer Science and Engineering
Shahajalal University of Science & Technology

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